

# RETAIL SALES ANALYSIS USING SQL

## DATA ANALYSIS PROJECT

### 1. INTRODUCTION

Retail businesses generate large amounts of data every day through customer purchases, product sales, and business operations. Analyzing this data can help organizations understand customer behavior, identify sales trends, and make better business decisions.

The purpose of this project is to analyze retail sales data using SQL and extract meaningful insights that can support business growth. The analysis was performed using the Sample Superstore dataset, which contains information related to orders, customers, products, sales, profit, and geographical regions.

In this project, SQL was used to answer important business questions related to sales performance, profitability, customer segments, and product demand. The analysis focuses on identifying top-performing product categories, high-value customers, profitable and loss-making products, and sales performance across different regions and states.

By performing this analysis, valuable insights were generated that can help businesses improve their sales strategies, better understand customer purchasing patterns, and make data-driven decisions. This project also demonstrates the practical application of SQL in solving real-world business problems and highlights the importance of data analysis in the retail industry.

### 2. OBJECTIVES

The main objectives of this project are:

- To calculate the overall sales revenue and profit generated by the company.
- To identify the top-performing product categories and sub-categories based on sales.
- To analyze sales performance across different regions and states.
- To identify high-value customers and customer segments.
- To determine the most profitable and loss-making products.
- To analyze customer purchasing behavior and product demand.
- To derive meaningful business insights that can support data-driven decision-making.

### 3. DATASET DESCRIPTION

The dataset used in this project is the **Sample Superstore dataset**, a widely used retail dataset for data analysis and business intelligence projects. The dataset contains transactional information related to customer orders, product sales, profit, and regional performance.

The dataset consists of **9,994 records** and **21 columns**, providing detailed information about various aspects of retail operations. It includes customer details, product information, sales figures, profit values, quantities sold, and geographical information such as region and state.

For this project, the dataset was imported into **SQLite** using **DB Browser for SQLite**. SQL queries were then executed to analyze the data and answer important business questions related to sales performance, customer behavior, profitability, and product demand.

Some of the key attributes used in this analysis include **Customer Name, Segment, Category, Sub-Category, Product Name, Sales, Quantity, Profit, Region, and State**. These attributes were used to generate insights that can support data-driven business decisions.

| Attribute     | Description                                                  |
|---------------|--------------------------------------------------------------|
| Row ID        | Unique identifier assigned to each row in the dataset        |
| Order ID      | Unique identifier for each customer order                    |
| Order Date    | Date on which the order was placed                           |
| Ship Date     | Date on which the order was shipped                          |
| Ship Mode     | Shipping method used for delivering the order                |
| Customer ID   | Unique identifier assigned to each customer                  |
| Customer Name | Name of the customer                                         |
| Segment       | Customer segment category (Consumer, Corporate, Home Office) |
| Country       | Country where the customer is located                        |
| City          | City where the order was placed                              |
| State         | State where the order was placed                             |
| Postal Code   | Postal code of the customer's location                       |
| Region        | Geographic sales region                                      |
| Product ID    | Unique identifier assigned to each product                   |
| Category      | Product category                                             |
| Sub-Category  | Product sub-category                                         |
| Product Name  | Name of the product                                          |
| Sales         | Revenue generated from product sales                         |
| Quantity      | Number of units sold                                         |
| Discount      | Discount applied to the product                              |
| Profit        | Profit earned from the sale                                  |

## 4. TOOLS AND TECHNOLOGIES

The following tools and technologies were used to perform data analysis and generate business insights in this project:

| Tool/Technology       | Purpose                                                           |
|-----------------------|-------------------------------------------------------------------|
| SQL                   | Used to query, filter, aggregate, and analyze the data            |
| SQLite                | Used as the relational database management system                 |
| DB Browser for SQLite | Used to import data, execute SQL queries, and manage the database |
| Microsoft Word        | Used to prepare and document the project report                   |
| GitHub                | Used to host and showcase the project portfolio                   |

## 5. SQL CONCEPTS

The following SQL concepts were applied throughout the project:

- **SELECT** – Used to retrieve specific data from the dataset.
- **GROUP BY** – Used to group records based on categories, regions, and customer segments.
- **ORDER BY** – Used to sort query results in ascending or descending order.
- **HAVING** – Used to filter grouped results based on aggregate conditions.
- **Aggregate Functions (SUM(), AVG())** – Used to calculate total sales, profit, and average values.
- **LIMIT** – Used to restrict the number of rows returned in the output.
- **Aliases (AS)** – Used to assign meaningful names to output columns.
- **Filtering and Sorting** – Used to analyze specific subsets of data.

## 6. BUSINESS ANALYSIS AND INSIGHTS

### 1. What is the total sales revenue generated by the company?

#### SQL Query:

```
SELECT ROUND(SUM(Sales),2) AS Total_Sales  
FROM superstore;
```

#### Insight:

The company generated a total sales revenue of **\$2,297,200.86**, indicating strong overall business performance across all regions and product categories.

## 2. What is the total profit earned by the company?

### SQL Query:

```
SELECT ROUND(SUM(Profit),2) AS Total_Profit
FROM superstore;
```

### Insight:

The company generated a total profit of **\$286,397.02** from total sales of **\$2.29 million**, indicating overall profitability across business operations. This suggests that the company is effectively converting sales into profit while maintaining sustainable business growth.

## 3. Which product categories generate the highest sales?

### SQL Query:

```
SELECT Category,
       ROUND(SUM(Sales),2) AS Total_Sales
FROM superstore
GROUP BY Category
ORDER BY Total_Sales DESC;
```

### Insight:

The **Technology** category generated the highest sales revenue (**\$836,154.03**), making it the company's top-performing category. **Furniture** ranked second with **\$741,999.80**, while **Office Supplies** contributed **\$719,047.03** in sales. This indicates that technology products are the primary drivers of overall business revenue.

## 4. Which sub-categories generate the highest sales?

### SQL Query:

```
SELECT "Sub-Category",
       ROUND(SUM(Sales),2) AS Total_Sales
FROM superstore
GROUP BY "Sub-Category"
ORDER BY Total_Sales DESC;
```

### Insight:

The **Phones** sub-category generated the highest sales revenue (**\$330,007.05**), closely followed by **Chairs** (**\$328,449.10**). Other top-performing sub-categories include **Storage** (**\$223,843.61**), **Tables** (**\$206,965.53**), and **Binders** (**\$203,412.73**). In contrast, sub-categories such as **Fasteners**, **Labels**, and **Envelopes** contributed the least to total sales. These findings indicate that technology-related and office furniture products are major contributors to the company's revenue.

## 5. Which regions generate the highest sales?

### SQL Query:

```
SELECT Region,
       ROUND(SUM(Sales),2) AS Total_Sales
FROM superstore
GROUP BY Region
ORDER BY Total_Sales DESC;
```

### Insight:

The **West** region generated the highest sales revenue, contributing **\$725,457.82** to total sales. The **East** region ranked second with **\$678,781.24**, followed by the **Central** region (**\$501,239.89**) and the **South** region (**\$391,721.91**). These results indicate that the West and East regions are the company's strongest markets and contribute significantly to overall business performance.

## 6. Who are the top 10 customers by sales?

### SQL Query:

```
SELECT "Customer Name",
       ROUND(SUM(Sales),2) AS Total_Sales
FROM superstore
GROUP BY "Customer Name"
ORDER BY Total_Sales DESC
LIMIT 10;
```

### Insight:

**Sean Miller** emerged as the highest-value customer, generating total sales of **\$25,043.05**. Other top customers include **Tamara Chand** (**\$19,052.22**), **Raymond Buch** (**\$15,117.34**), and **Tom Ashbrook** (**\$14,595.62**). Identifying high-value customers enables the company to develop targeted marketing strategies, strengthen customer relationships, and improve customer retention efforts.

## 7. Which products are generating losses for the company?

### SQL Query:

```
SELECT "Product Name",  
       ROUND(SUM(Profit),2) AS Total_Profit  
FROM superstore  
GROUP BY "Product Name"  
HAVING Total_Profit < 0  
ORDER BY Total_Profit ASC  
LIMIT 10;
```

### Insight:

Several products were identified as significant contributors to business losses. The **Cubify CubeX 3D Printer Double Head Print** generated the highest loss of **\$8,879.97**, followed by the **Lexmark MX611dhe Monochrome Laser Printer** with a loss of **\$4,589.97**. Other loss-making products include conference tables, binding systems, and videoconferencing equipment. These findings suggest that the company should review pricing strategies, discount policies, and inventory management practices for these products to improve overall profitability.

## 8. Which customer segment contributes the highest revenue?

### SQL Query:

```
SELECT Segment,  
       ROUND(SUM(Sales),2) AS Total_Sales  
FROM superstore  
GROUP BY Segment  
ORDER BY Total_Sales DESC;
```

### Insight:

The **Consumer** segment contributed the highest revenue, generating **\$1,161,401.34** in sales. The **Corporate** segment ranked second with **\$706,146.37**, while the **Home Office** segment contributed **\$429,653.15**. These findings indicate that individual consumers represent the company's largest customer base and are the primary drivers of overall sales revenue.

## 9. Which states generate the highest sales?

### SQL Query:

```
SELECT State,  
       ROUND(SUM(Sales),2) AS Total_Sales
```

```
FROM superstore
GROUP BY State
ORDER BY Total_Sales DESC
LIMIT 10;
```

**Insight:**

**California** generated the highest sales revenue, contributing **\$457,687.63**, followed by **New York** with **\$310,876.27**. **Texas, Washington,** and **Pennsylvania** also emerged as key contributors to overall sales. The strong performance of California and New York indicates that these states represent major markets for the company and should continue to be strategic focus areas for business growth and customer engagement.

**10. Which products generate the highest profit?**

**SQL Query:**

```
SELECT "Product Name",
       ROUND(SUM(Profit),2) AS Total_Profit
FROM superstore
GROUP BY "Product Name"
ORDER BY Total_Profit DESC
LIMIT 10;
```

**Insight:**

The **Canon imageCLASS 2200 Advanced Copier** emerged as the most profitable product, generating a total profit of **\$25,199.93**. Other highly profitable products include the **Fellowes PB500 Electric Punch Plastic Comb Binding Machine (\$7,753.04)** and the **Hewlett Packard LaserJet 3310 Copier (\$6,983.88)**. The analysis indicates that copiers, printers, and office equipment significantly contribute to the company's profitability and should be prioritized in future sales and marketing strategies.

**11. Which states generate the highest profit?**

**SQL Query:**

```
SELECT State,
       ROUND(SUM(Profit),2) AS Total_Profit
FROM superstore
GROUP BY State
ORDER BY Total_Profit DESC
LIMIT 10;
```

**Insight:**

**California** generated the highest profit, contributing **\$76,381.39**, closely followed by **New York** with **\$74,038.55**. Other highly profitable states include **Washington (\$33,402.65)** and **Michigan (\$24,463.19)**. These findings indicate that California and New York are not only strong revenue-generating markets but also major contributors to overall business profitability. Focusing on these high-performing states can support sustained business growth and strategic decision-making.

## 12. What is the average sales amount for each customer segment?

### SQL Query:

```
SELECT Segment,
       ROUND(AVG(Sales),2) AS Average_Sales
FROM superstore
GROUP BY Segment
ORDER BY Average_Sales DESC;
```

### Insight:

The **Home Office** segment recorded the highest average sales amount per order at **\$240.97**, followed by the **Corporate** segment with **\$233.82** and the **Consumer** segment with **\$223.73**. Although the Consumer segment generated the highest total revenue, the Home Office segment demonstrated the highest spending behavior on average, indicating greater purchasing value per transaction.

## 13. Which are the top 10 products sold in the highest quantities?

### SQL Query:

```
SELECT "Product Name",
       SUM(Quantity) AS Total_Quantity
FROM superstore
GROUP BY "Product Name"
ORDER BY Total_Quantity DESC
LIMIT 10;
```

### Insight:

**Staples** was the most frequently purchased product, with a total quantity of **215 units sold**. Other highly demanded products include **Staple envelope (170 units)** and **Easy-staple paper (150 units)**. The analysis reveals that office supply products dominate the list of top-selling items by quantity, indicating consistent demand for everyday office essentials. Understanding customer purchasing patterns can help the company optimize inventory management and improve product availability.

## DATA VISUALIZATIONS

To better communicate the findings, key business insights were visualized using charts created in Microsoft Excel. These visualizations provide a clear representation of sales performance across product categories, regions, and states. The charts help in identifying important business trends and support better decision-making.

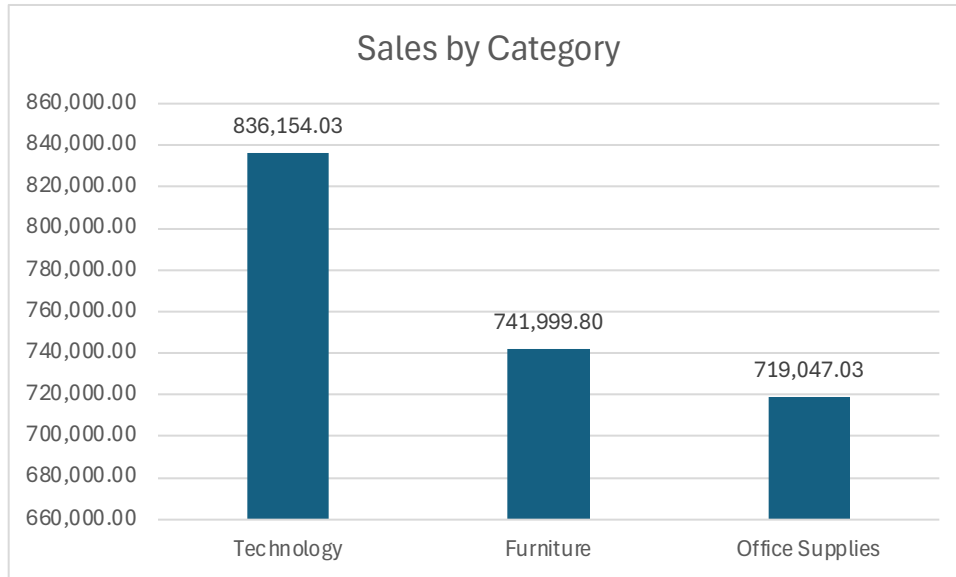


Figure 1: Sales by Category

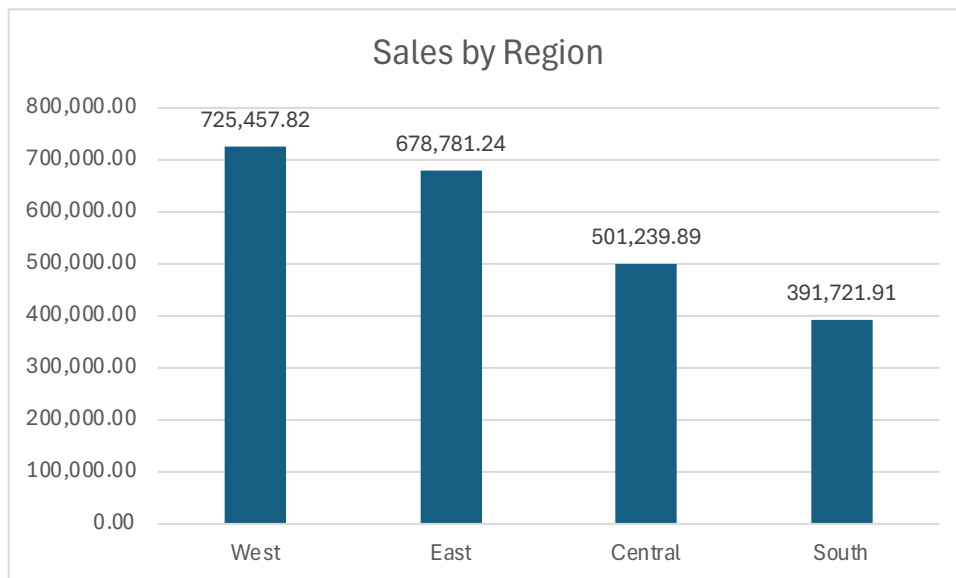


Figure 2: Sales by Region

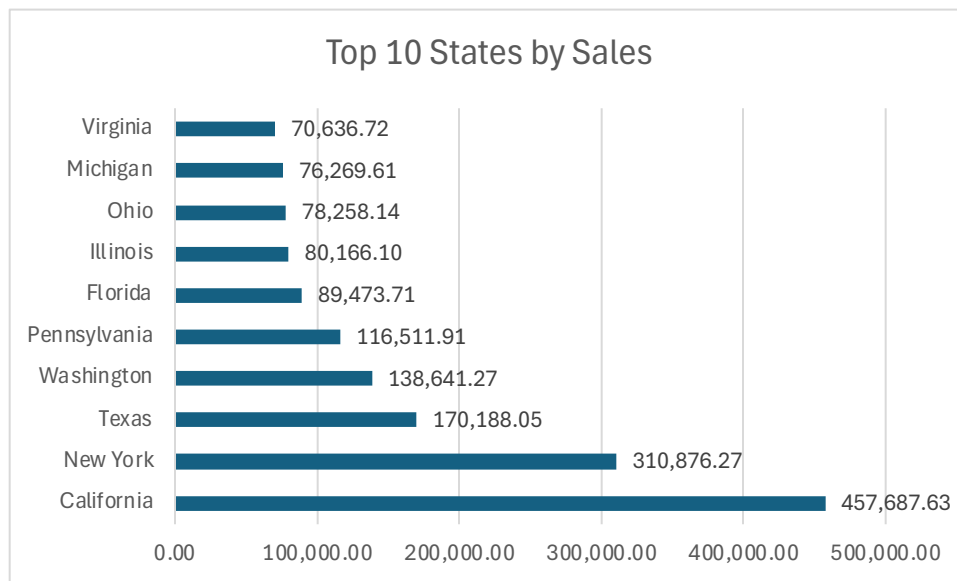


Figure 3: Top 10 States by Sales

## 7. CONCLUSION

This project analyzed retail sales data using SQL to identify important business insights related to sales performance, profitability, customer behavior, and product demand. The analysis revealed that the Technology category generated the highest sales revenue, while California and New York emerged as the top-performing states in terms of both sales and profit.

The project also identified high-value customers, profitable products, and products that contributed to business losses. Additionally, customer segment analysis showed that the Consumer segment generated the highest revenue, whereas the Home Office segment recorded the highest average sales per order.

Overall, this project demonstrates how SQL can be effectively used to analyze business data and support data-driven decision-making. The insights generated from this analysis can help organizations improve sales strategies, optimize inventory management, and enhance overall business performance.

